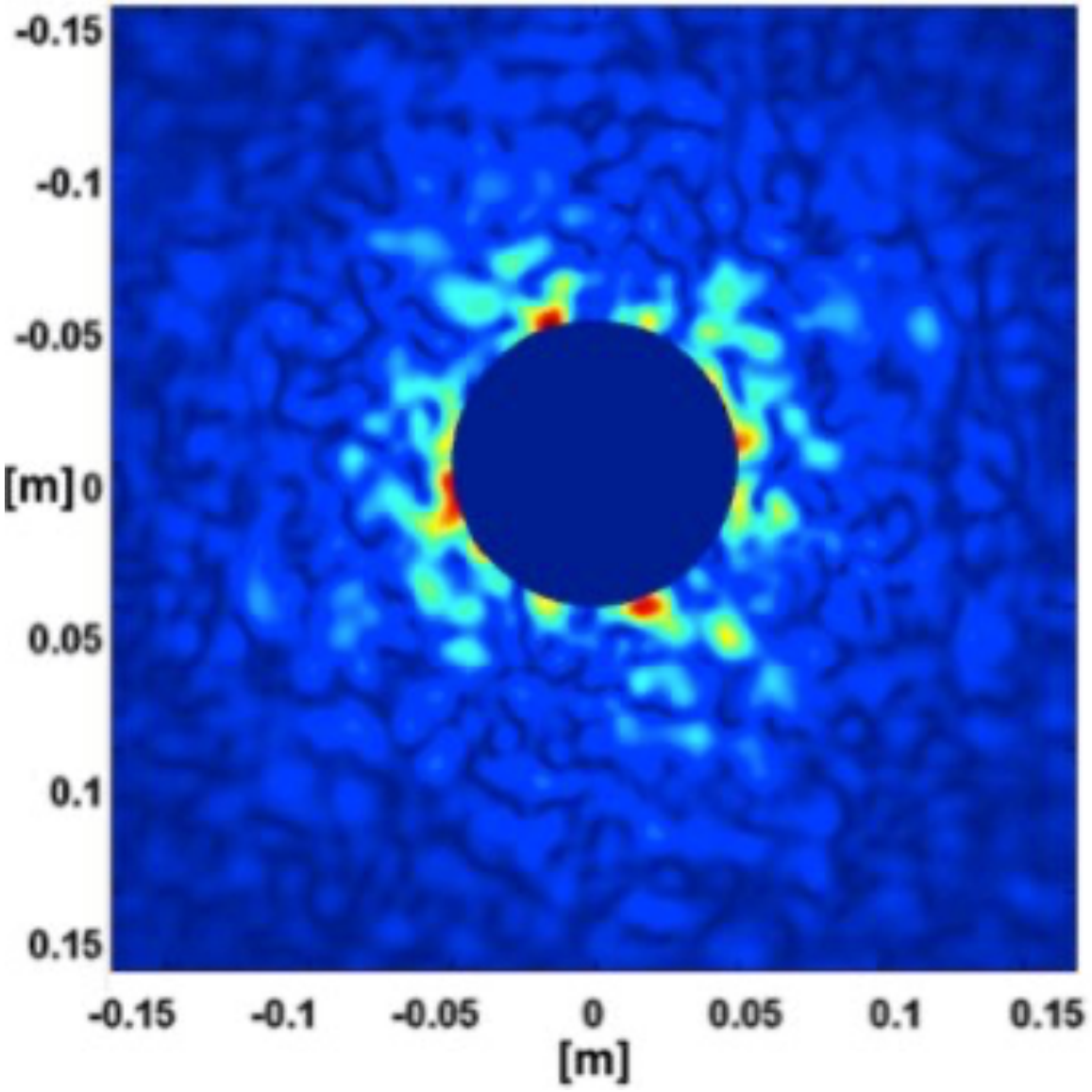
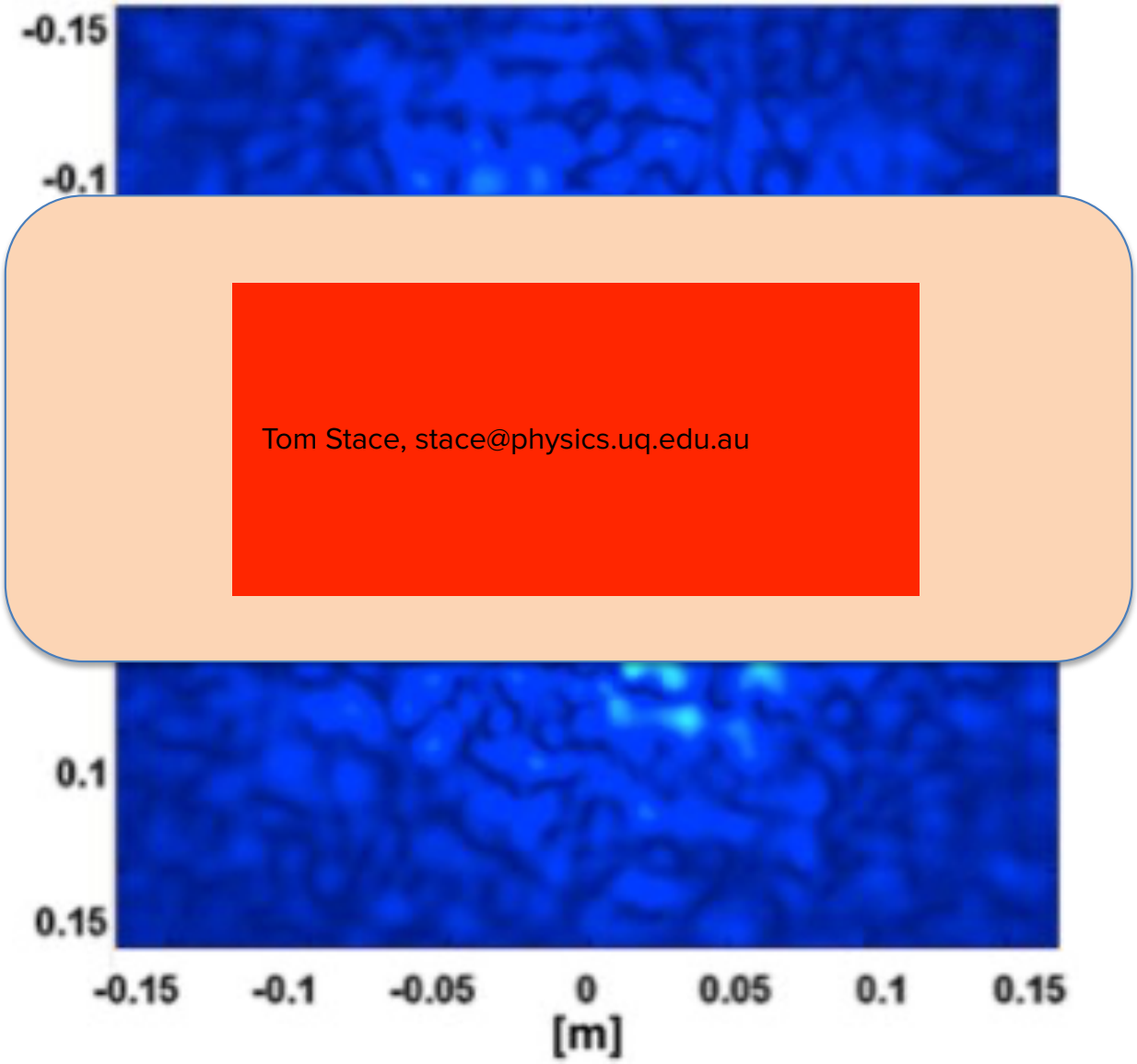
PHYS7045 – Quantum Noise, Decoherence and Errors



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Tom Stace, stace@physics.uq.edu.au

PHYS7045 – Quantum Noise, Decoherence and Errors

1. Open Quantum Systems
2. Quantum Channels
3. Quantum Noise, Decoherence and Errors
4. Physical Quantum Channels
5. Quantum Langevin Equations
6. The Master Equation
7. Conditional Quantum Dynamics

PHYS7045 – Quantum Noise, Decoherence and Errors

Objectives:

1. Learn physical principles of Open Quantum Systems
2. Calculate physically relevant quantities for Open Quantum Systems
3. Study experimental examples
4. Explore various methods for modelling Open Quantum Systems

PHYS7045 – Quantum Noise, Decoherence and Errors

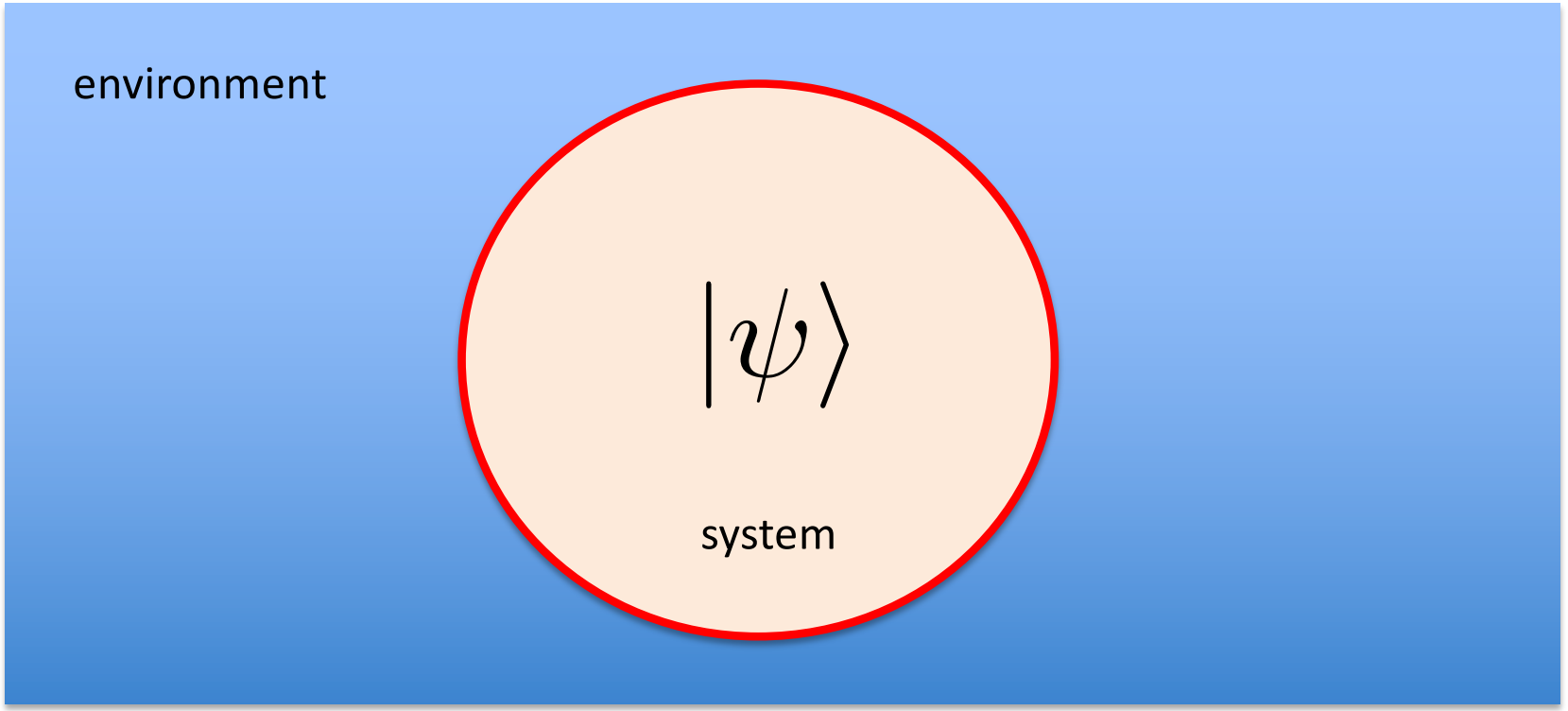
- 1 assignment due:

- Overheads & additional notes on blackboard
- Other references:
  - Nielsen and Chuang, *Quantum Computation and Quantum Information*
  - Bachor and Ralph, *A Guide to Experiments in Quantum Optics*
  - Wiseman and Milburn, *Quantum Measurement and Control*

Open Quantum Systems

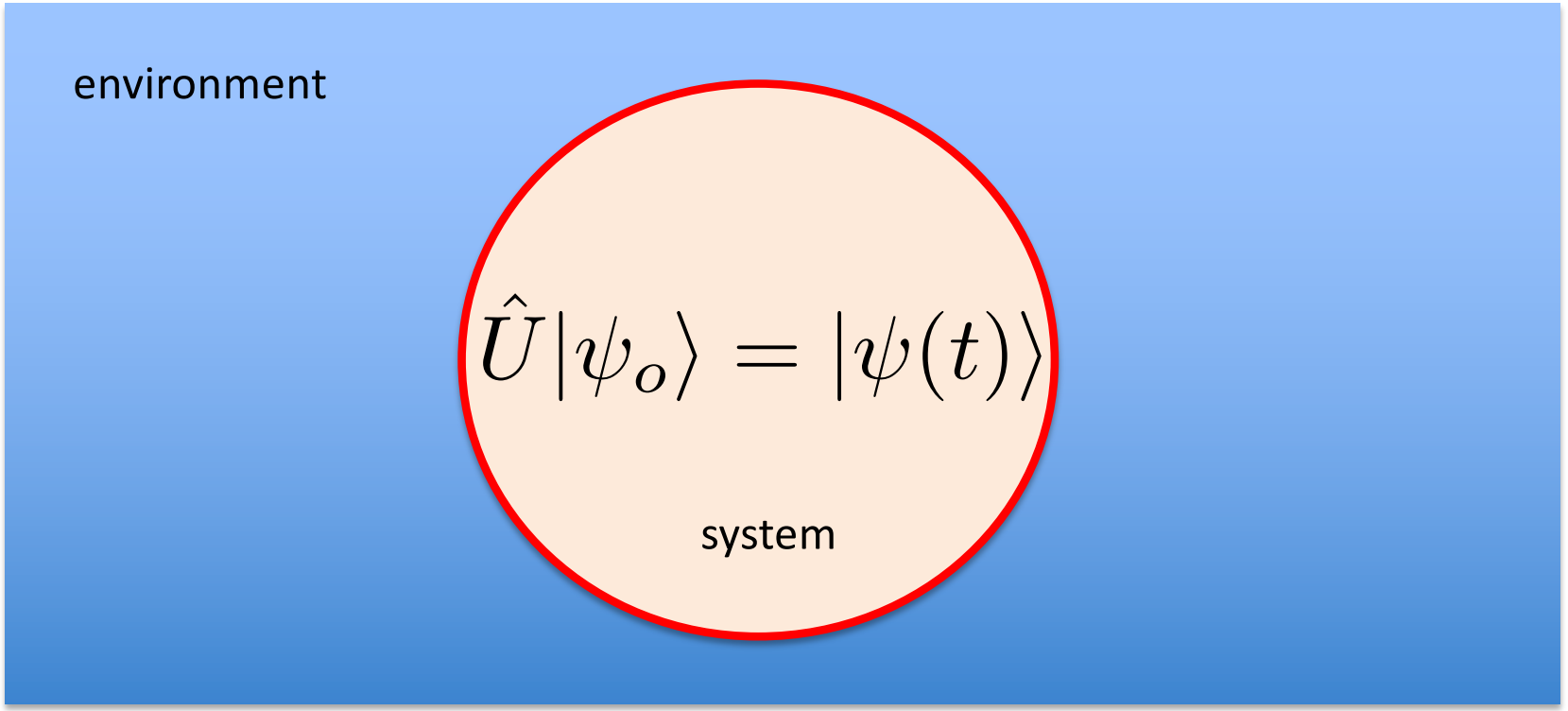
- Open vs closed quantum systems
- Examples of open quantum systems
  - atoms
  - artificial atoms
  - optics
- The quantum technology compromise

Open vs closed quantum systems



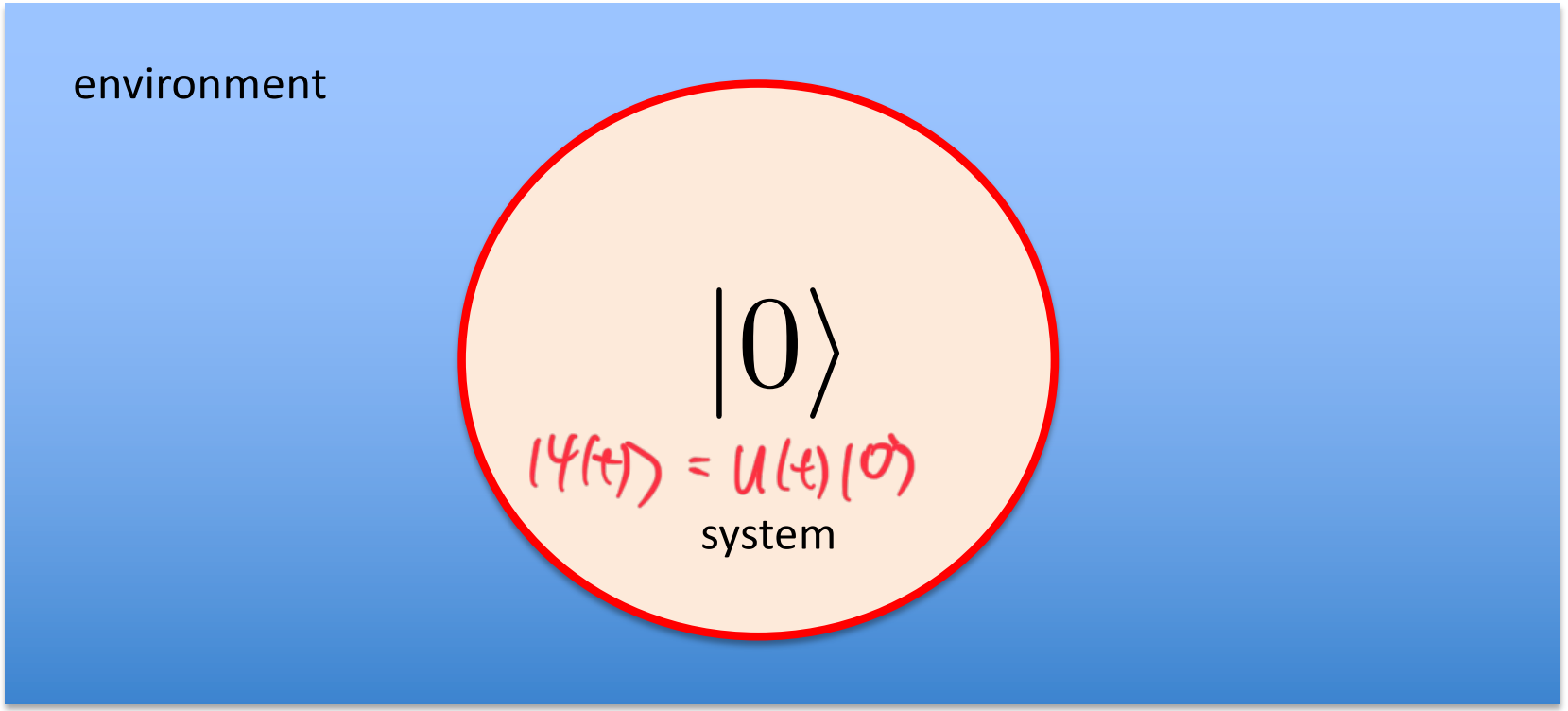
- A closed quantum system is isolated from the environment

Open vs closed quantum systems



- A closed quantum system is isolated from the environment
  - it evolves unitarily

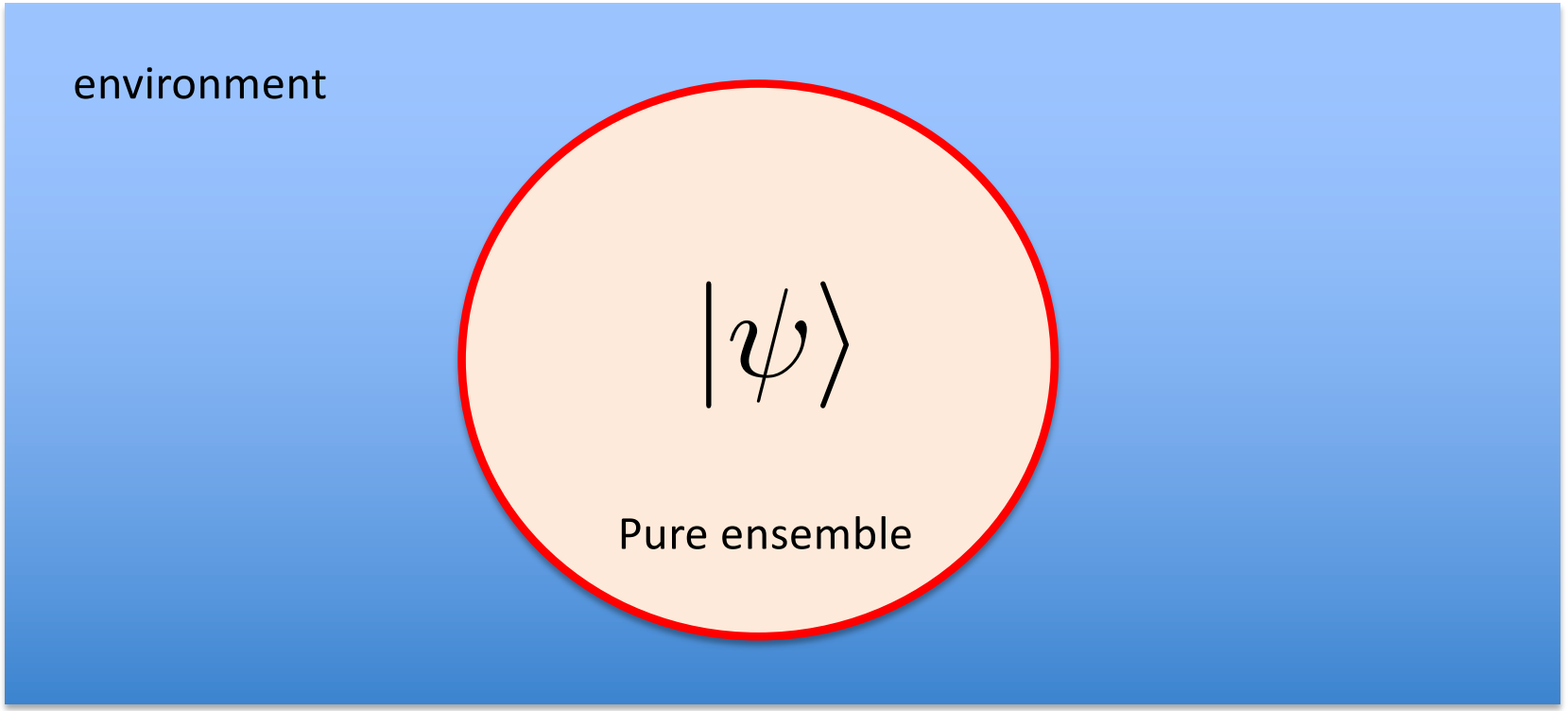
Open vs closed quantum systems



- A closed quantum system is isolated from the environment
  - it evolves unitarily
  - there is always some measurement basis for which it will give a definite result

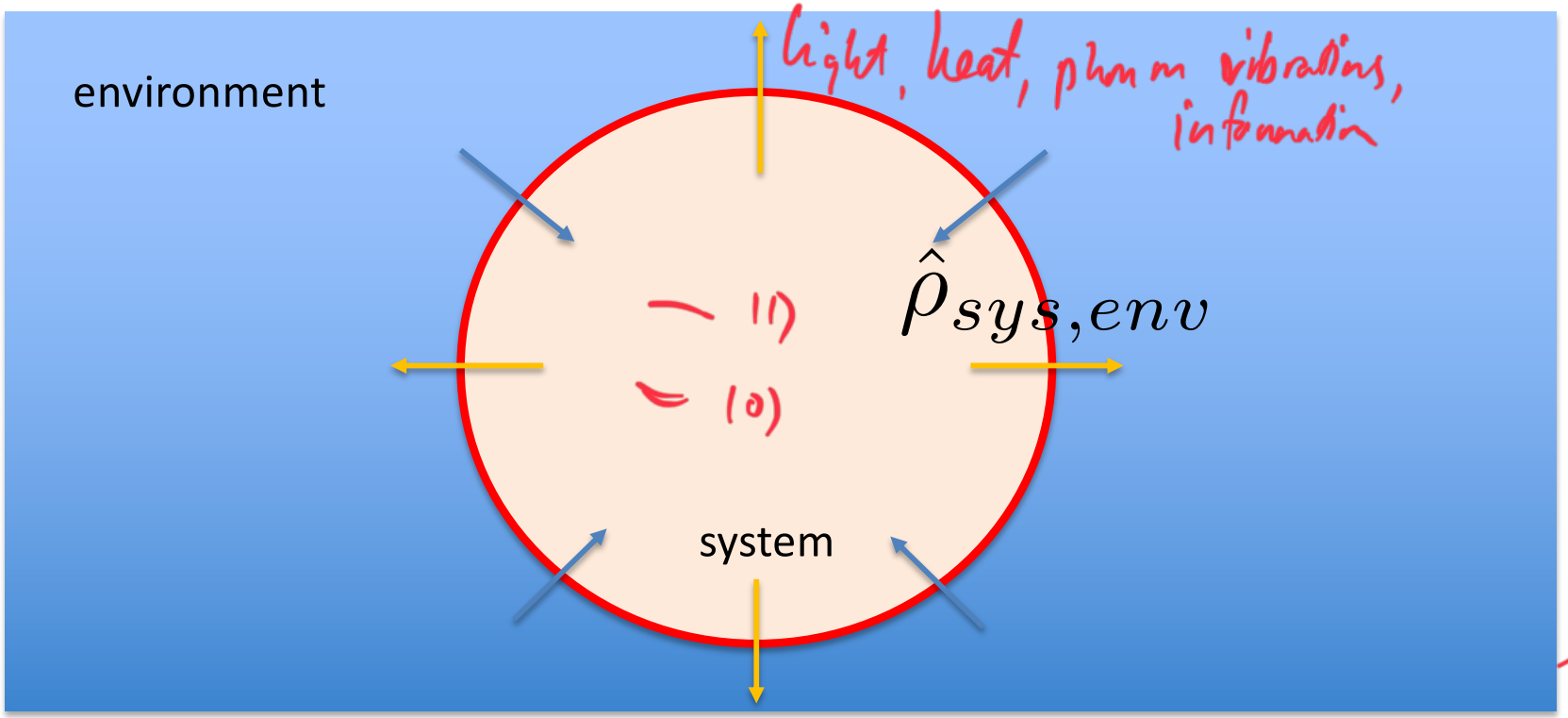
$U(t) = e^{-iHt}$ $\{|0\rangle, U(t)|0\rangle, |1\rangle, \ldots\}$

Open vs closed quantum systems



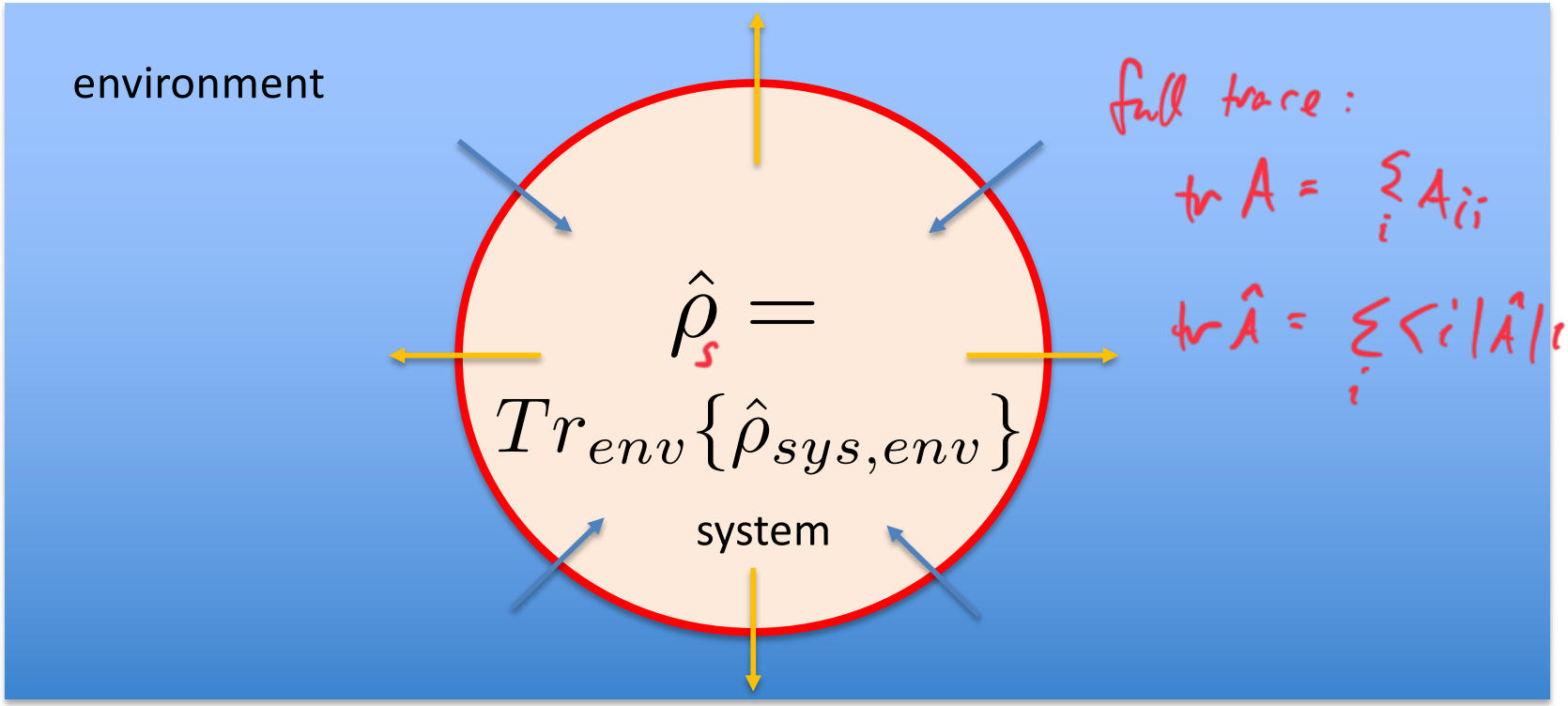
- A closed quantum system is isolated from the environment
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Open vs closed quantum systems



- An open quantum system interacts with the environment

Open vs closed quantum systems



full trace:

$\mathrm{tr}\, A = \sum_i A_{ii}$

$\mathrm{tr}\, \hat{A} = \sum_i \langle i|\hat{A}|i\rangle$

- An open quantum system interacts with the environment

$\hat{\rho}_s$ is the "reduced" density matrix for the system

$\hat{\rho}_s \in \mathcal{H}_s$

Partial Trace of an operator $\rho_{S,E} \in \mathcal{H}_S \otimes \mathcal{H}_E$

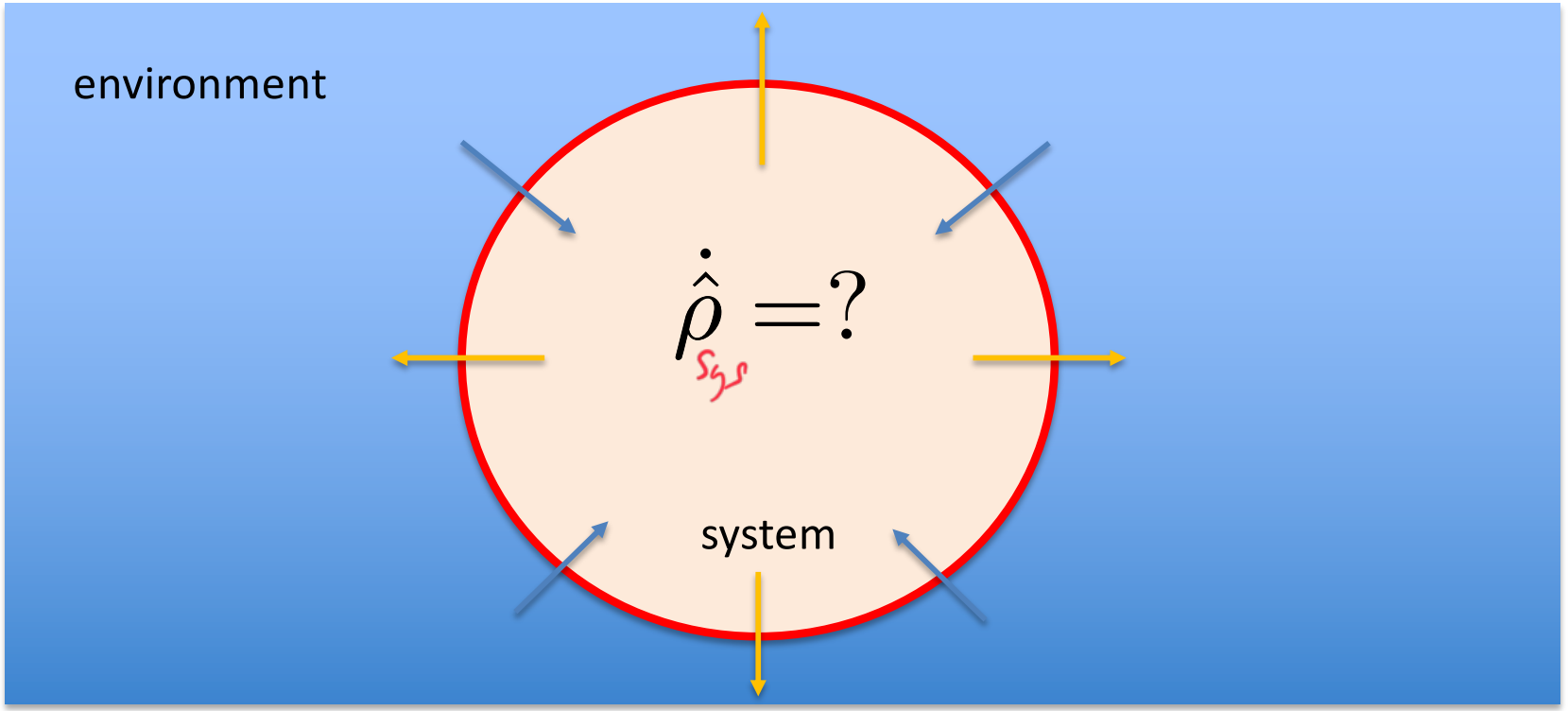
$\mathcal{H}_S = \text{span}\{|0\rangle_S, |1\rangle_S, |2\rangle_S, \dots\}$ $\dim \mathcal{H}_S = s$

$\mathcal{H}_E = \text{span}\{|0\rangle_E, |1\rangle_E, |2\rangle_E, \dots\}$ $\dim \mathcal{H}_E = e$

$\rho_S = \text{tr}_E \rho_{S,E} = \sum_k \langle k|_E \rho_{S,E} |k\rangle_E \in \mathcal{H}_S$

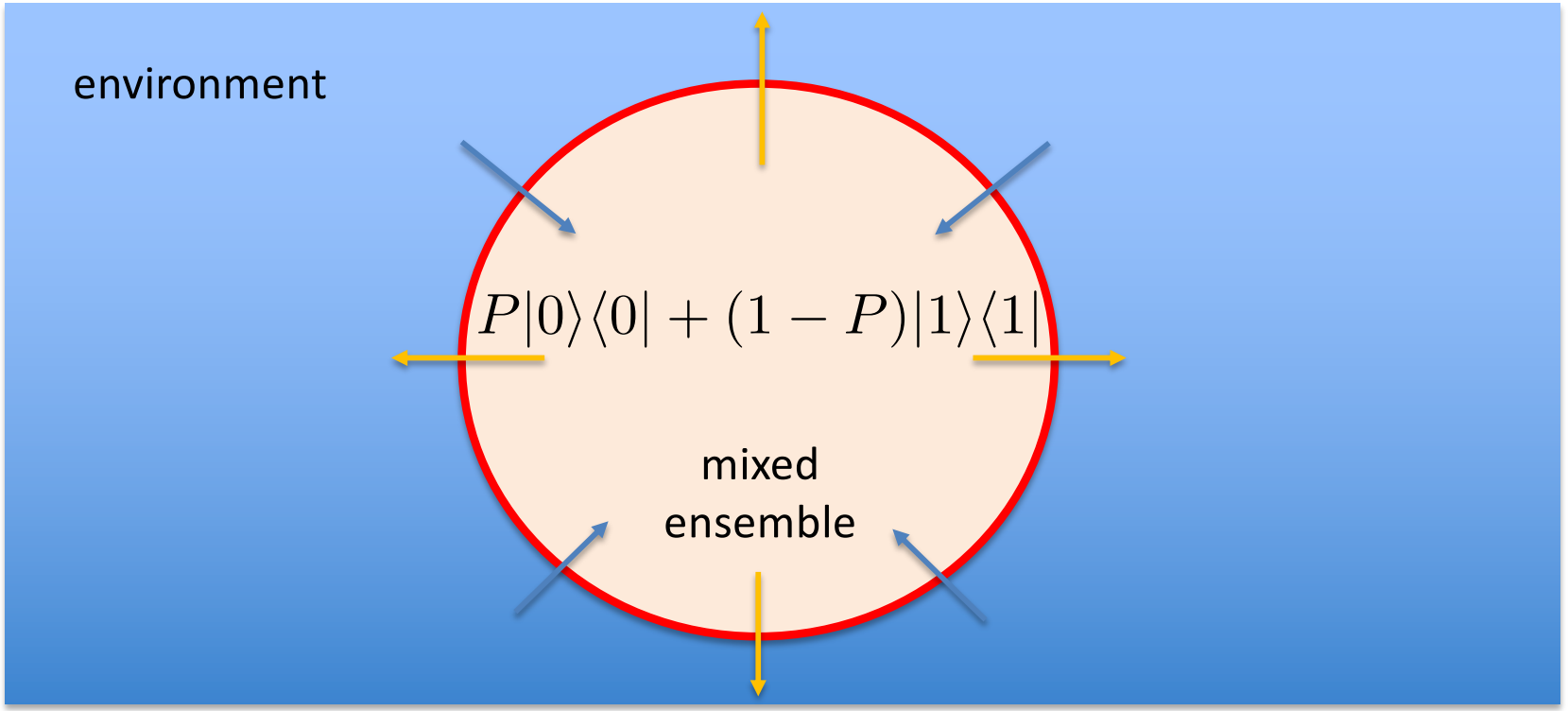
$s \times s = \dim$ $\dim = se \times se$

Open vs closed quantum systems



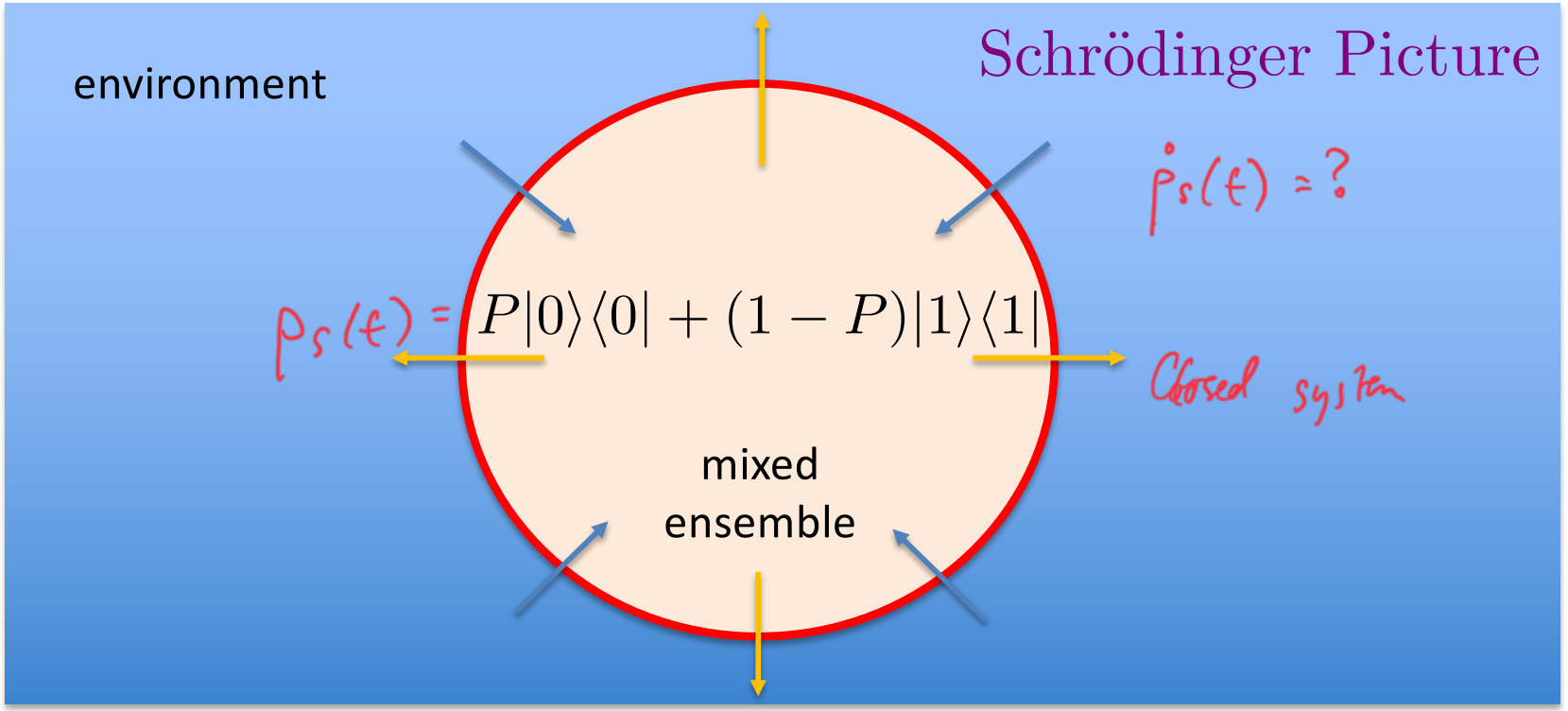
- An open quantum system interacts with the environment
  - it doesn't evolve unitarily

Open vs closed quantum systems



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Open vs closed quantum systems



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Time evolution of a density matrix for a closed system

- environment writes the initial state
- closed unitary evolution

$\dot{\rho} \stackrel{?}{=}$

$|\dot{\psi}\rangle = -iH|\psi\rangle$

Schrodinger equation

$\rho(0) = p|0\rangle\langle 0| + (1-p)|1\rangle\langle 1|$

Special case of pure state evolution:

$\rho_0 = |\psi_0\rangle\langle\psi_0| \Rightarrow \rho(t) = |\psi(t)\rangle\langle\psi(t)|$

$\Rightarrow \dot{\rho}(t) = \left(\frac{d}{dt}|\psi(t)\rangle\right)\langle\psi(t)| + |\psi(t)\rangle\left(\frac{d}{dt}\langle\psi(t)|\right)$

$= |\dot{\psi}\rangle\langle\psi| + |\psi\rangle\langle\dot{\psi}|$

$$= -iH|\psi\rangle\langle\psi| + |\psi\rangle\langle\psi|iH$$

$$= -i\left(H|\psi\rangle\langle\psi| - |\psi\rangle\langle\psi|H\right)$$

$$= -i\left(H\rho(t) - \rho(t)H\right)$$

$$= -i[H, \rho(t)] = \frac{d}{dt}\rho(t)$$ - von Neumann equation

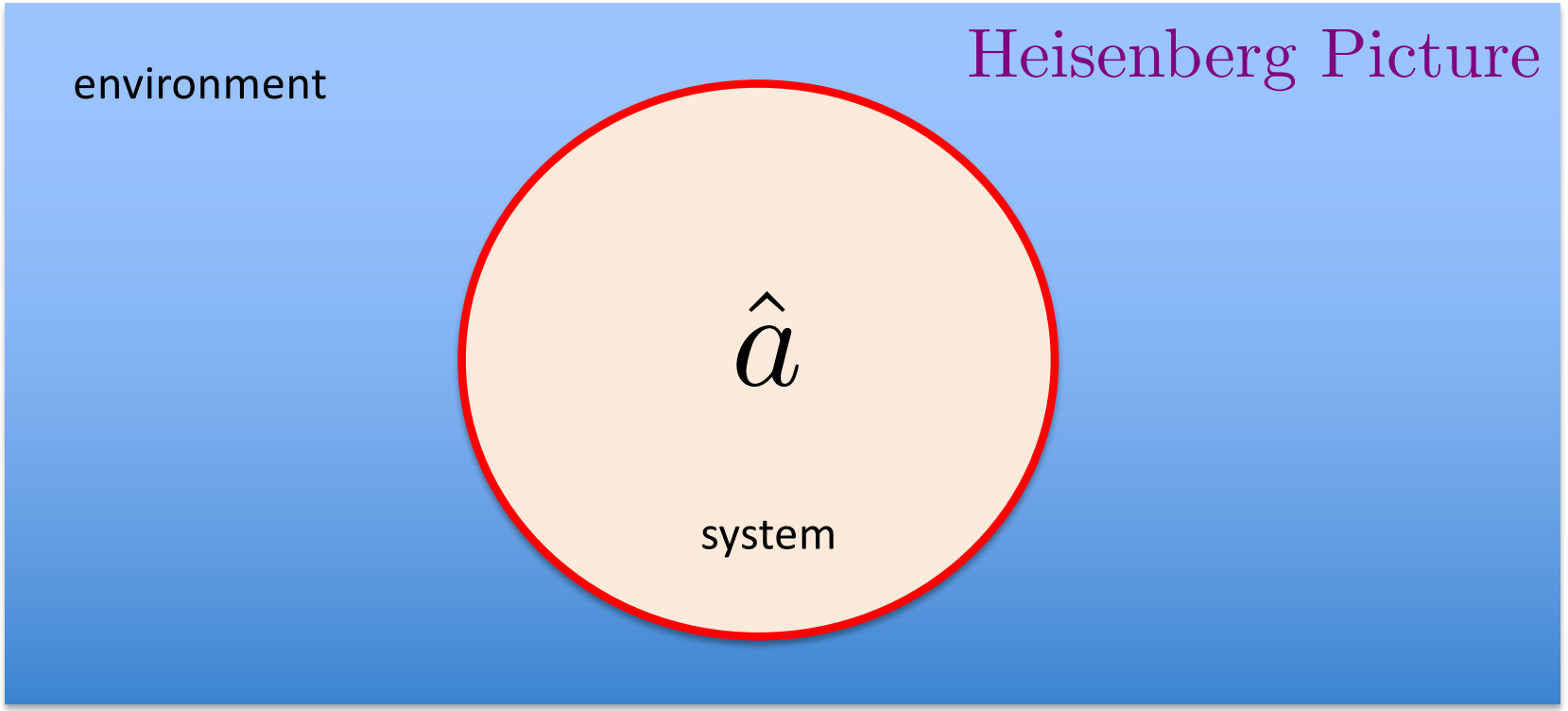
If the initial state is a mixture of pure states:

$$\rho(0) = P\,|0\rangle\langle 0| + (1-P)\,|1\rangle\langle 1|$$
$$= P\,\rho_0(0) + (1-P)\,\rho_1(0)$$

$$\dot{\rho}(t) = P\left(-i[H, \rho_0(t)]\right) + (1-P)\left(-i[H, \rho_1(t)]\right)$$

$$= -i[H, P\rho_0(t) + (1-P)\rho_1(t)] = -i[H, \rho(t)]$$

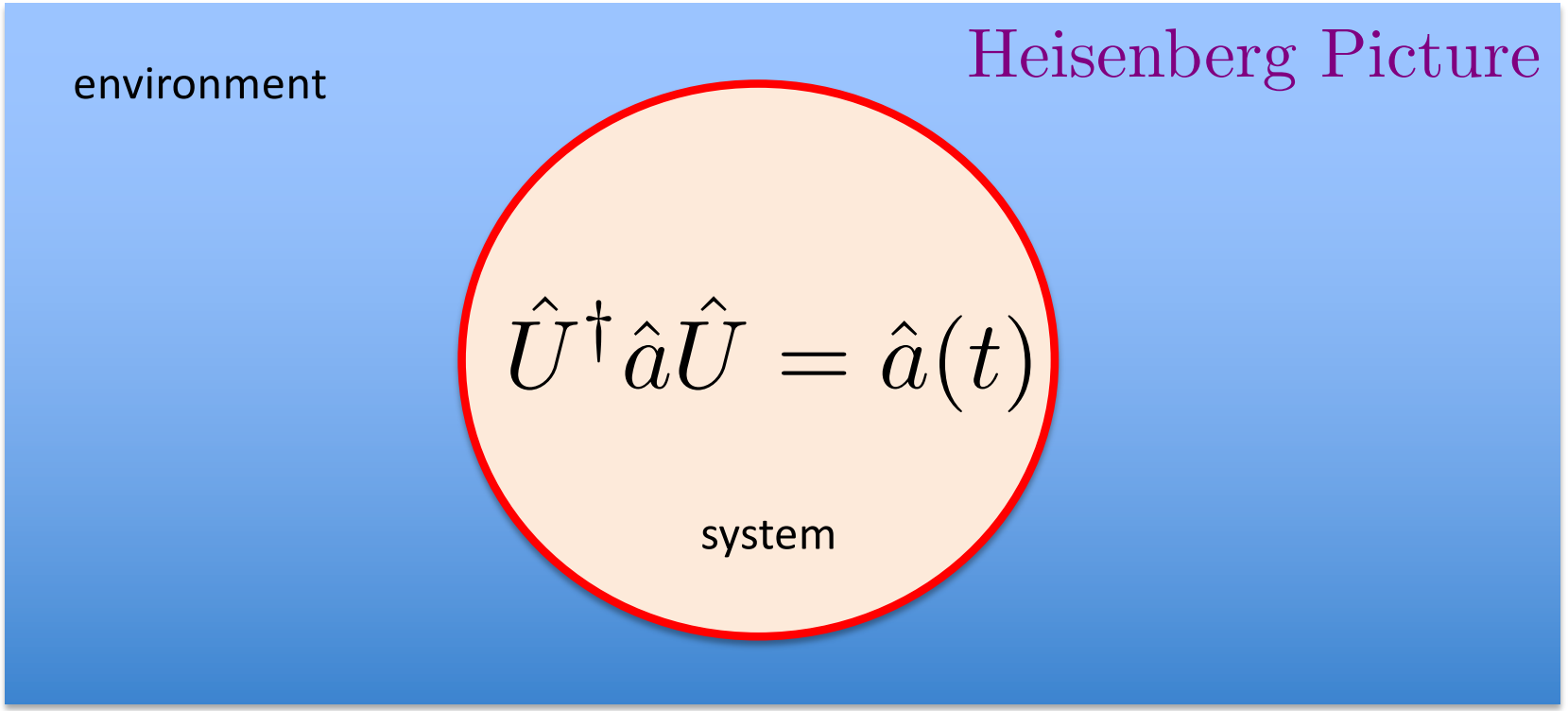
Open vs closed quantum systems



- A closed quantum system is isolated from the environment

$$\frac{d\hat{a}}{dt} = i[H, \hat{a}]$$

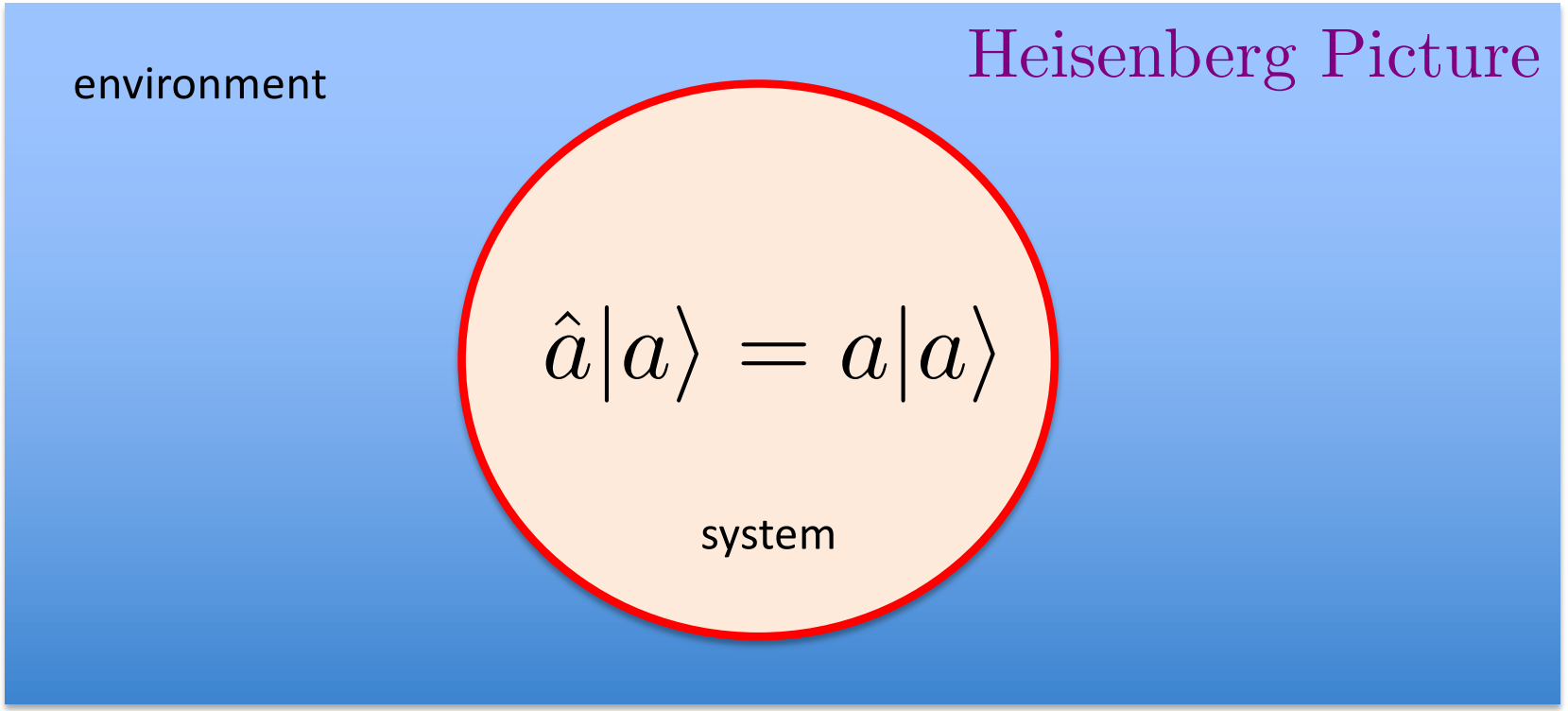
Open vs closed quantum systems



- A closed quantum system is isolated from the environment
  - it evolves unitarily

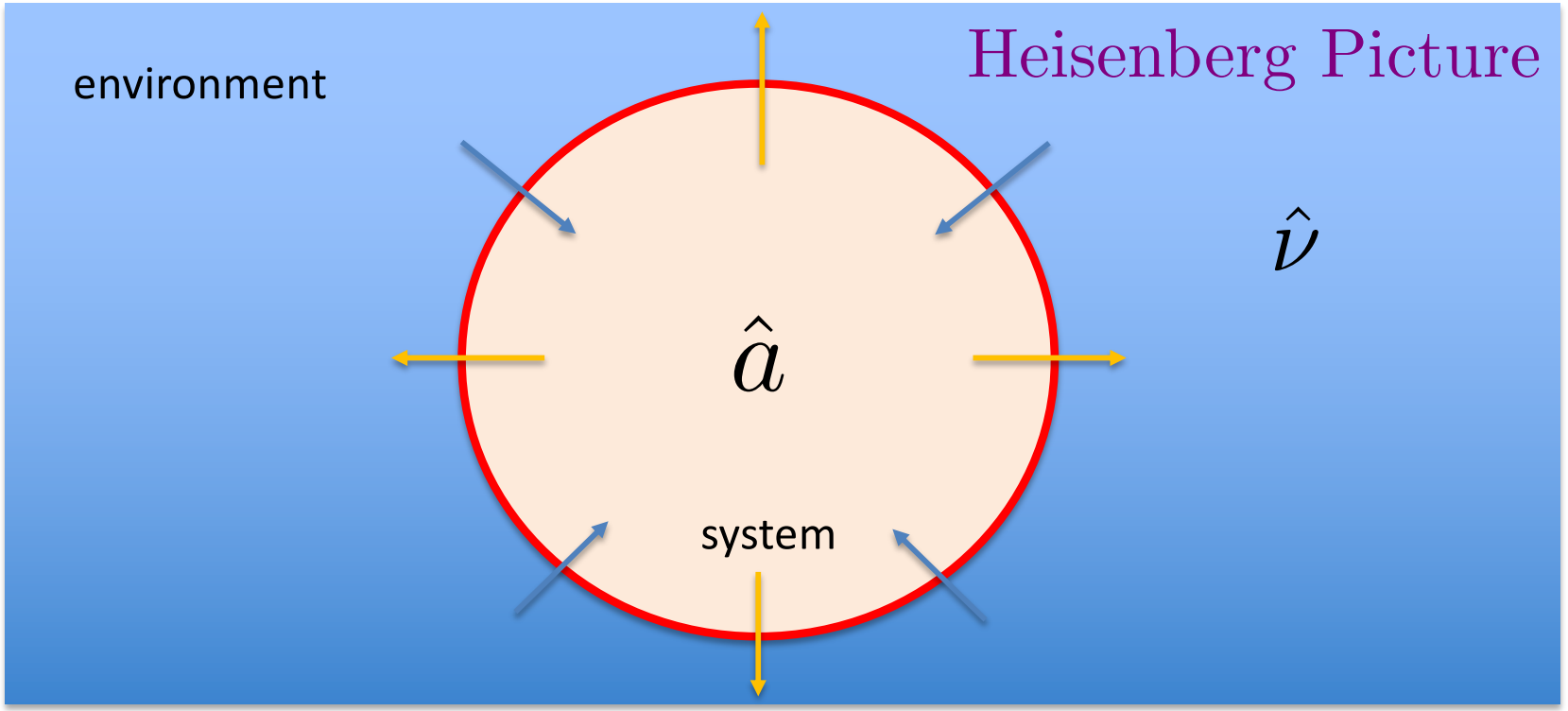
$U = e^{-iHt}$

Open vs closed quantum systems



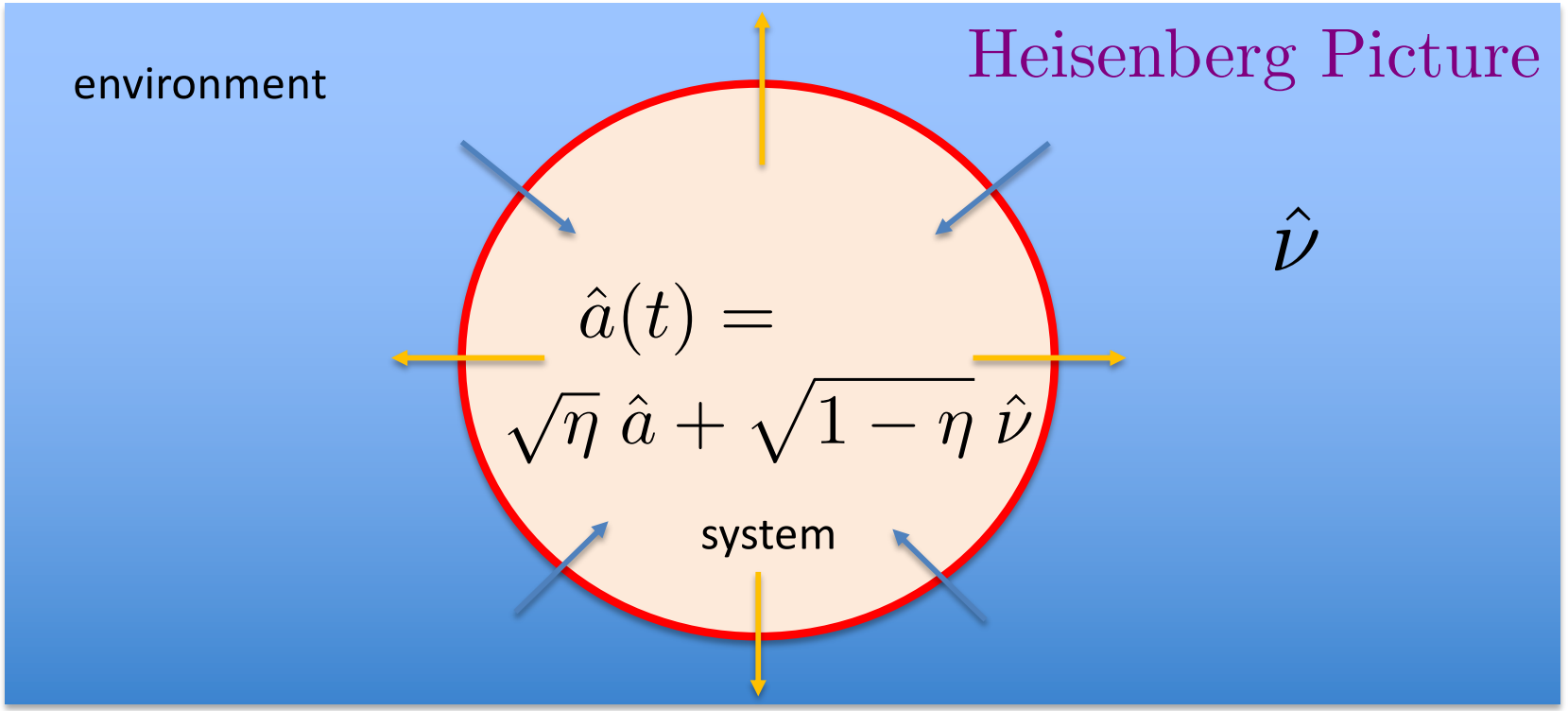
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Open vs closed quantum systems



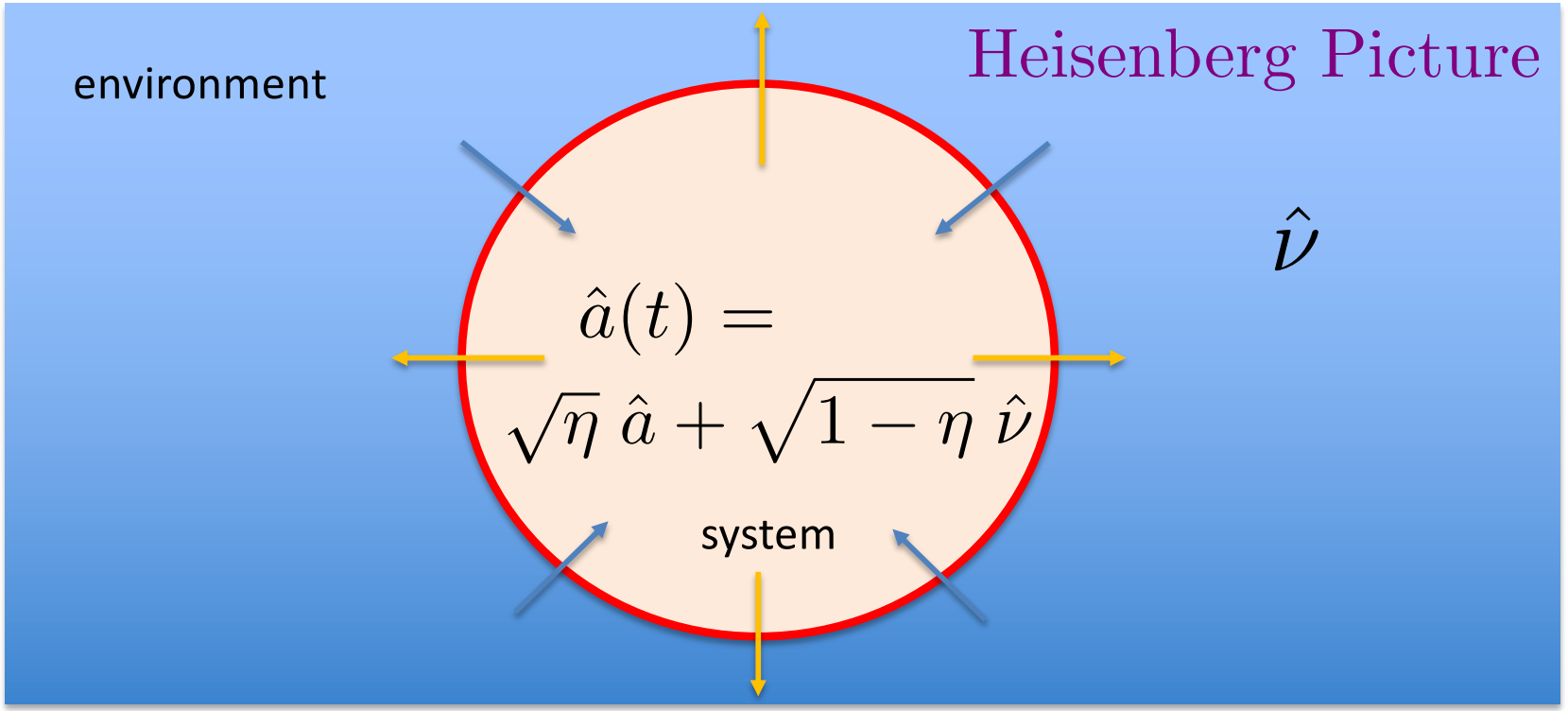
- An open quantum system interacts with the environment

Open vs closed quantum systems



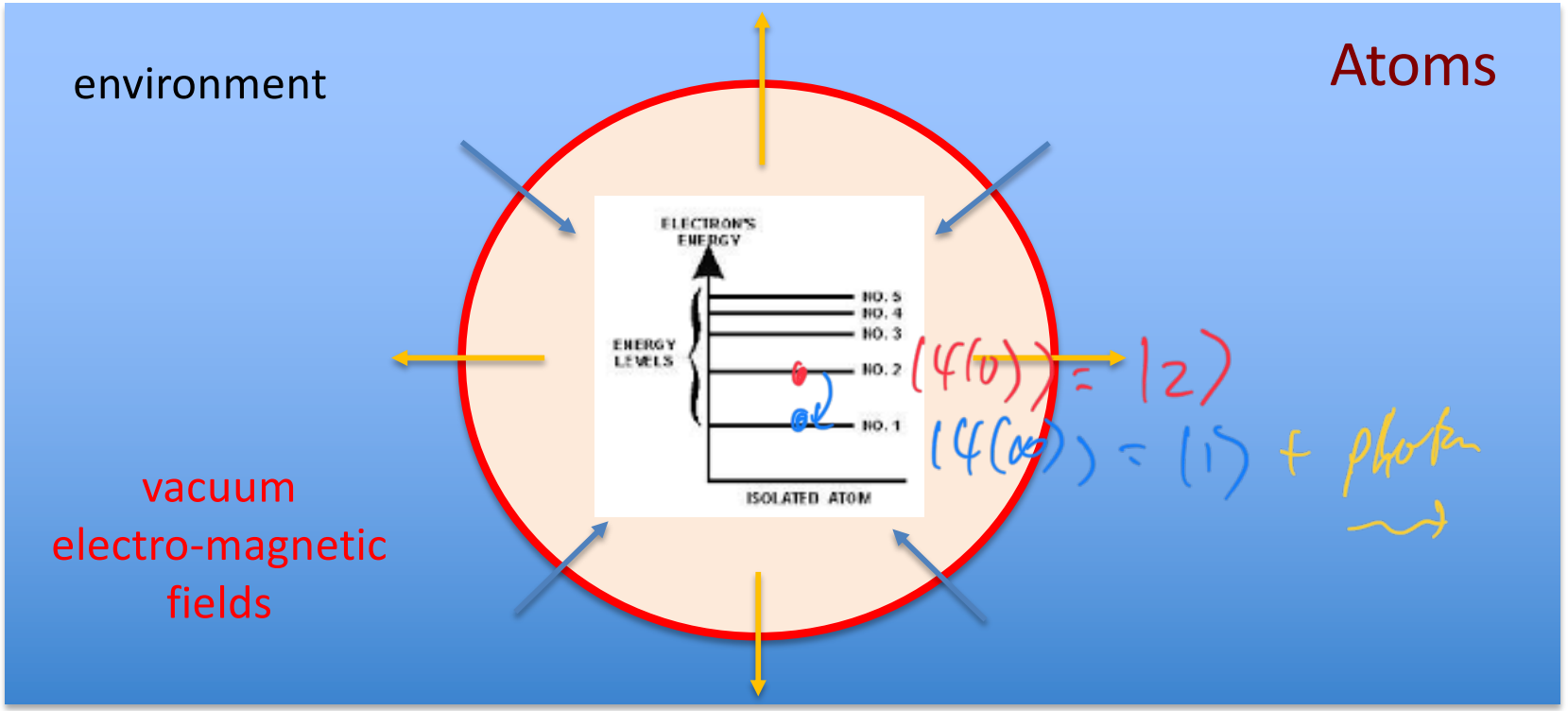
- An open quantum system interacts with the environment
  - it doesn't evolve unitarily

Open vs closed quantum systems



- An open quantum system interacts with the environment
  - it doesn't evolve unitarily
  - there doesn't exist a measurement basis for which it will give a definite result

Examples of open quantum systems



- In free-space a solitary atom interacts with the E/M environment
  - excited states spontaneously decay to lower energy states

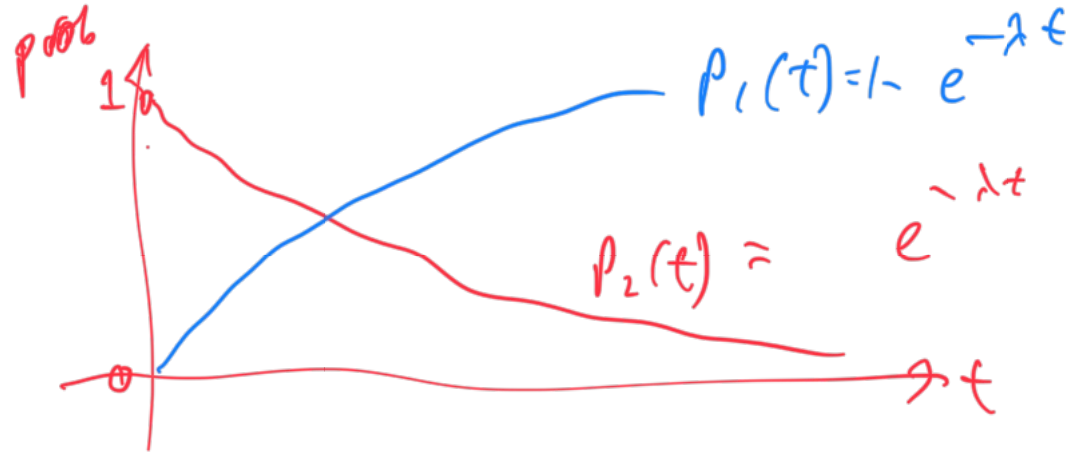
$\hat{\rho}_S(0) = |2\rangle\langle 2|$ $\quad$ $\hat{\rho}_S(t) = \rho_2(t)\,|2\rangle\langle 2| + \rho_1(t)\,|1\rangle\langle 1|$

where $\rho_1(t) + \rho_2(t) = 1$ (assuming only relaxation).

We would expect exponential decay of the excited state

- also exponential evolution of the ground state

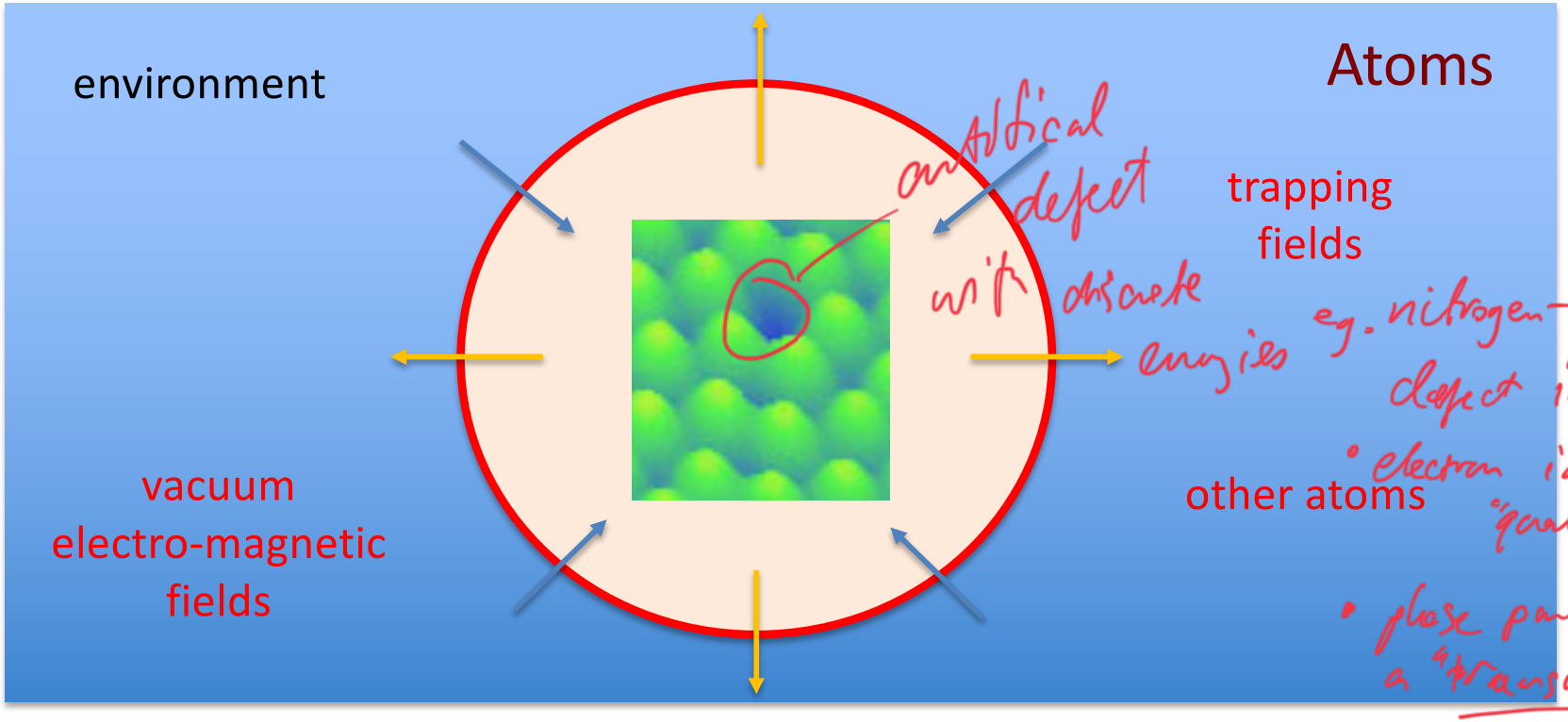
guess:



what equation of motion for $\rho(t)$ gives this behavior?

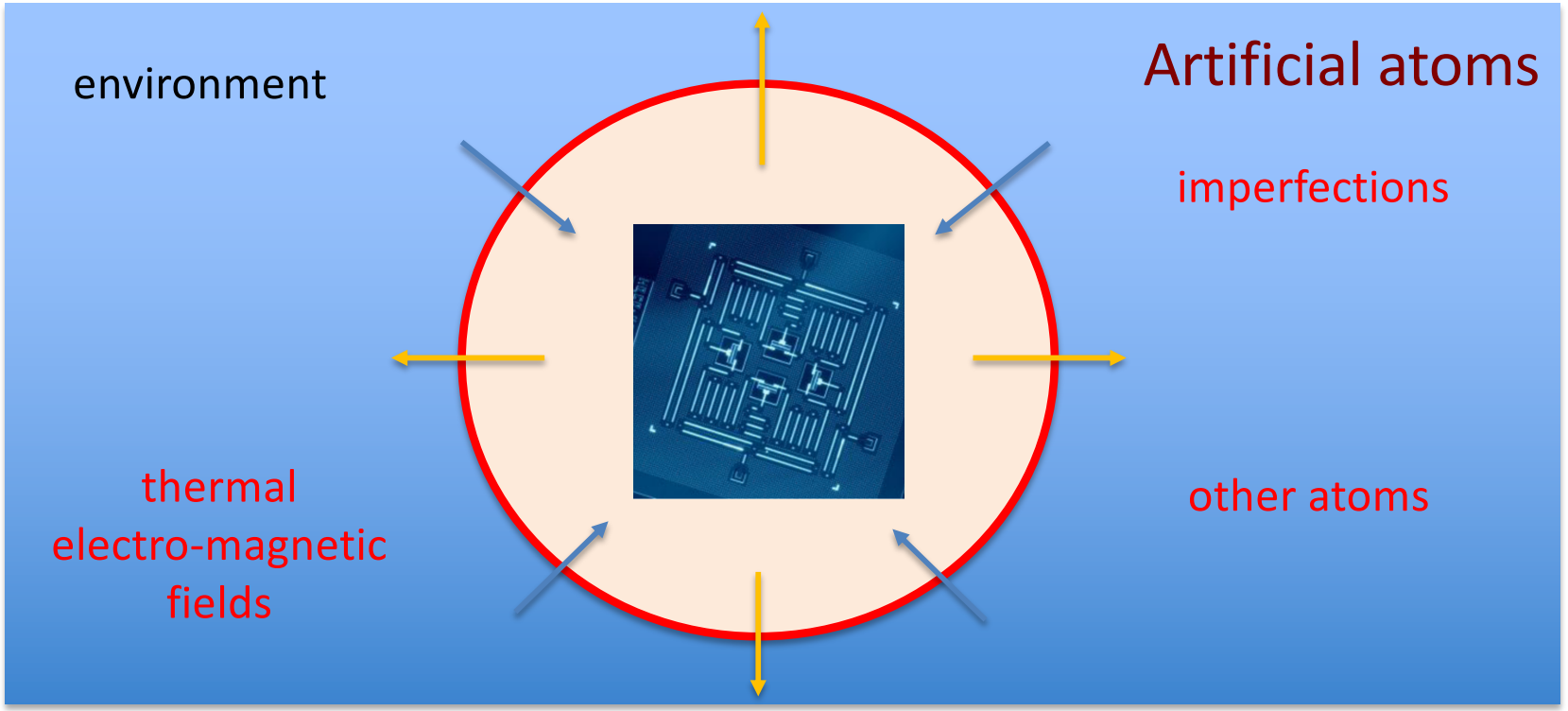
- Lindblad master equation, generalizes the von Neumann equation.

Examples of open quantum systems



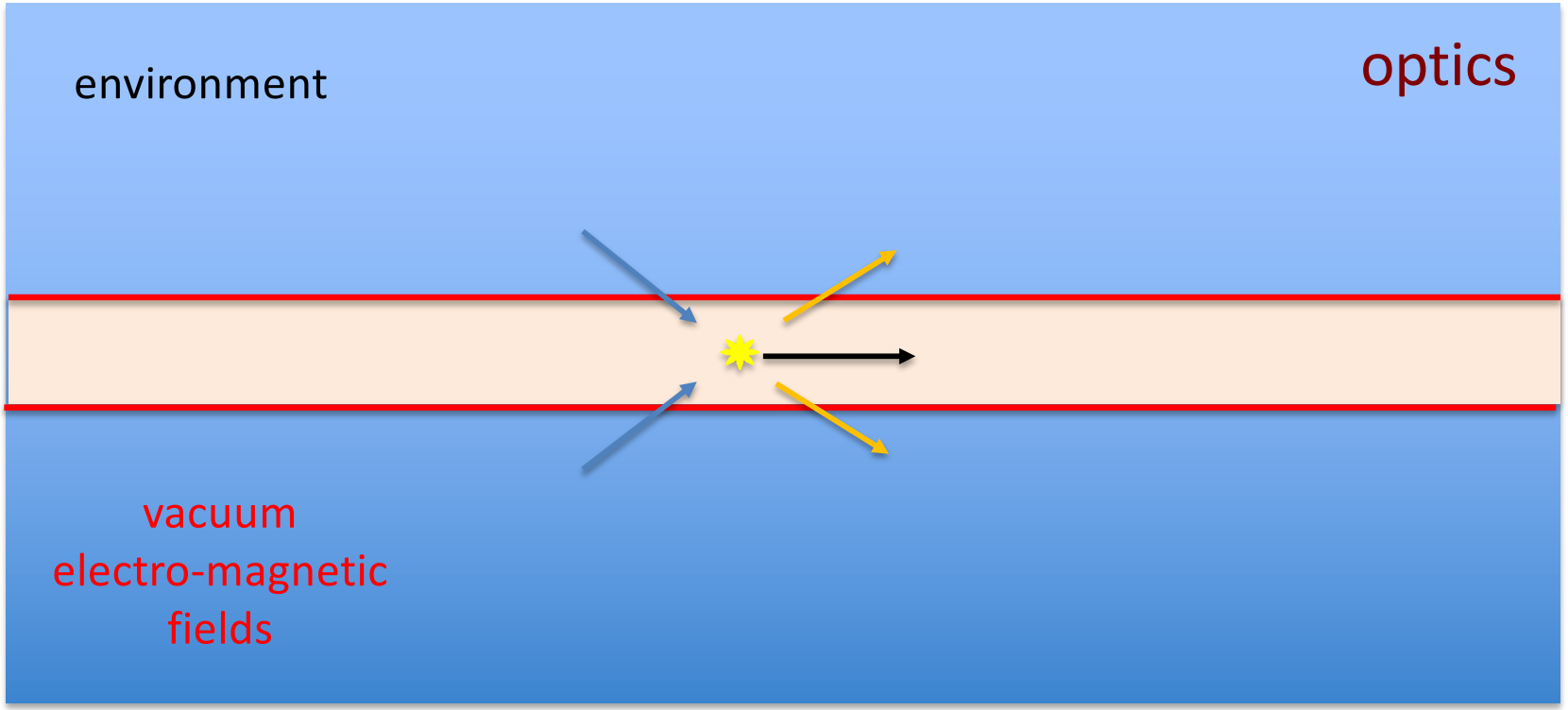
- Trapped atoms interact with the E/M environment plus other nearby atoms plus the trapping fields
  - excited states spontaneously decay to lower energy states
  - the coherence between energy levels can decay rapidly

Examples of open quantum systems



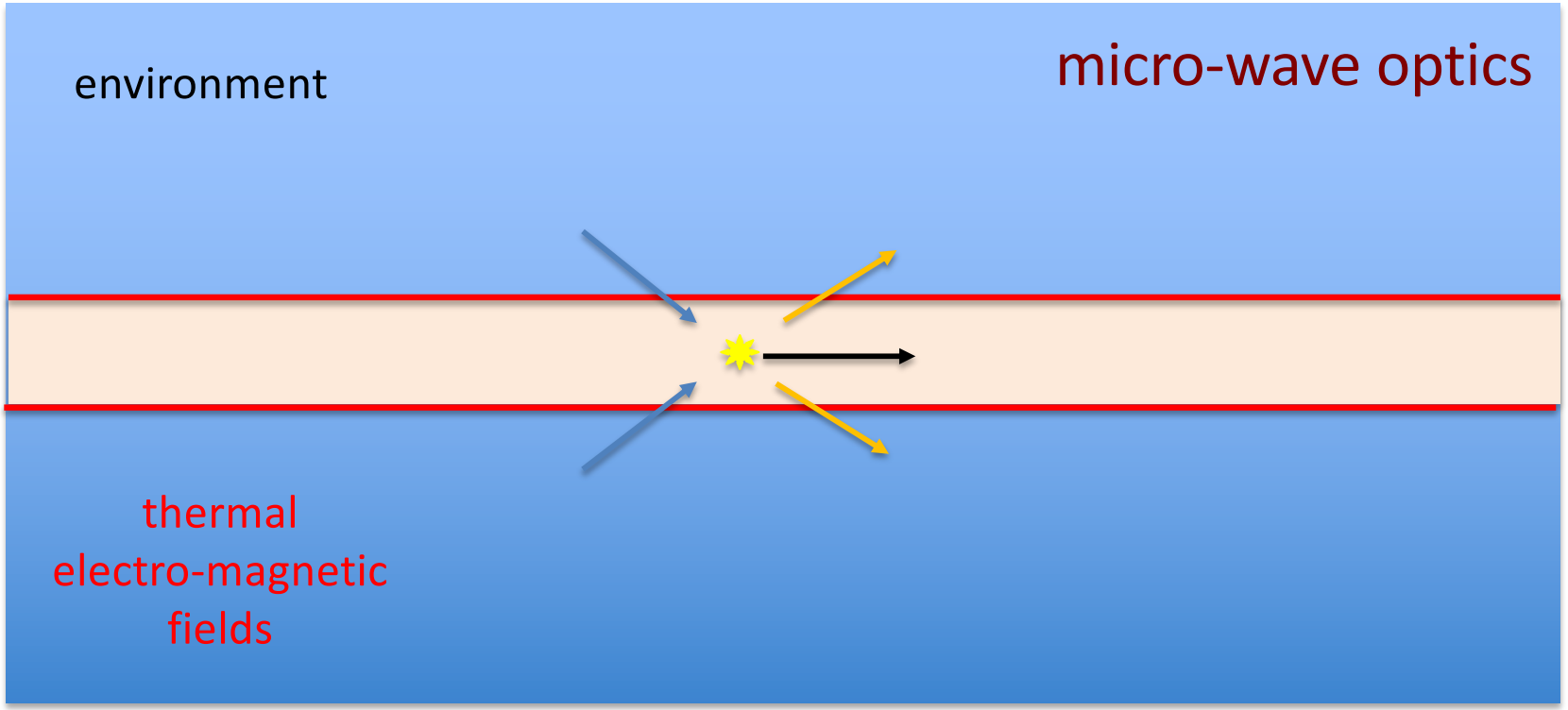
- Artificial atoms interact with the (thermal) E/M environment plus other nearby atoms plus the other imperfections
  - excited states spontaneously decay to lower energy states
  - the coherence between energy levels can decay rapidly

Examples of open quantum systems



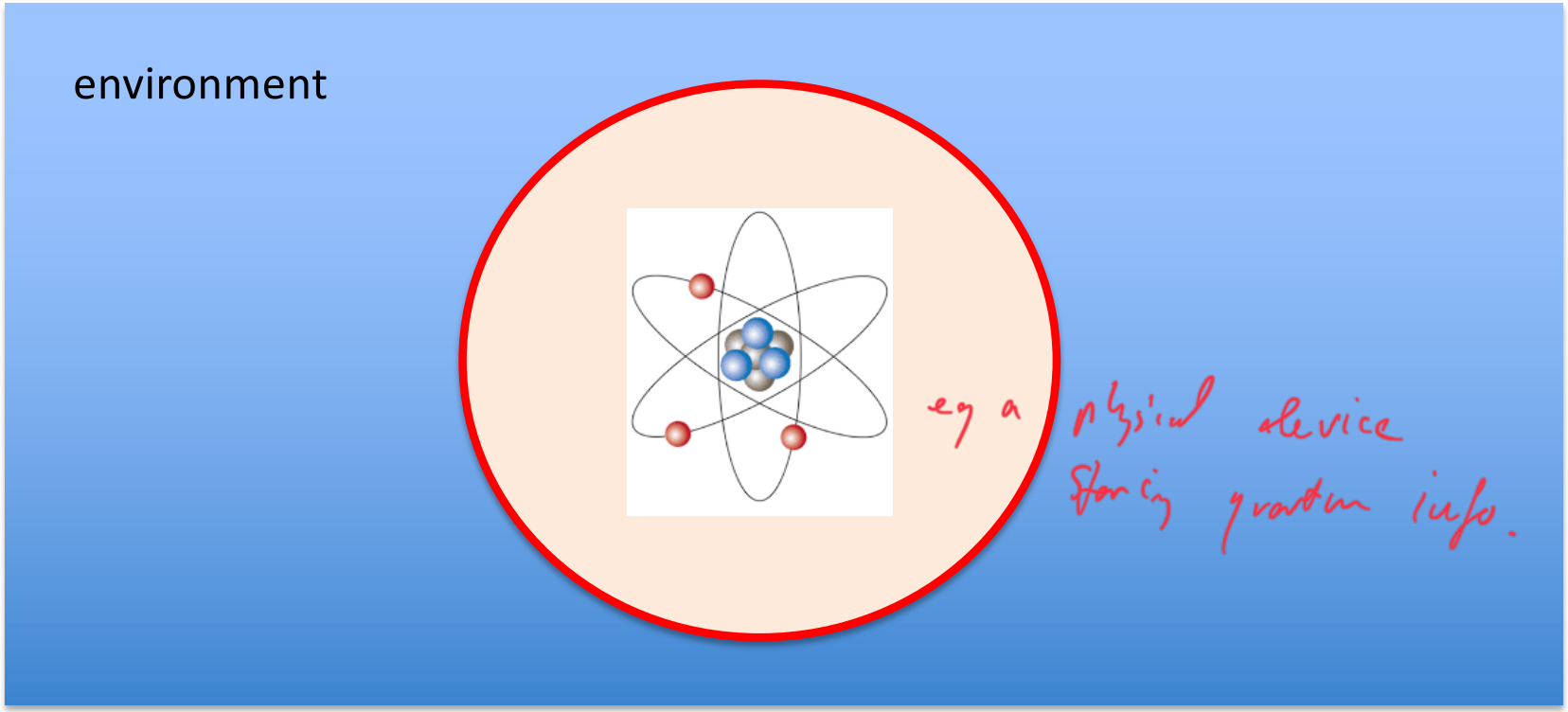
- Photons interact with the E/M environment via:
  - diffraction losses in free space
  - scattering losses due to guiding optics, fibres, lenses, etc
  - plus reduction of coherence due to fluctuations in refractive index, etc

Examples of open quantum systems



- Photons interact with the E/M environment via:
  - diffraction losses in free space
  - scattering losses due to guiding optics, fibres, lenses, etc
  - plus reduction of coherence due to fluctuations in refractive index, etc

The quantum technology compromise



- is this the ideal situation?
  - want to protect arbitrary quantum states ✓ superpositions of
  - want to manipulate quantum states ✗

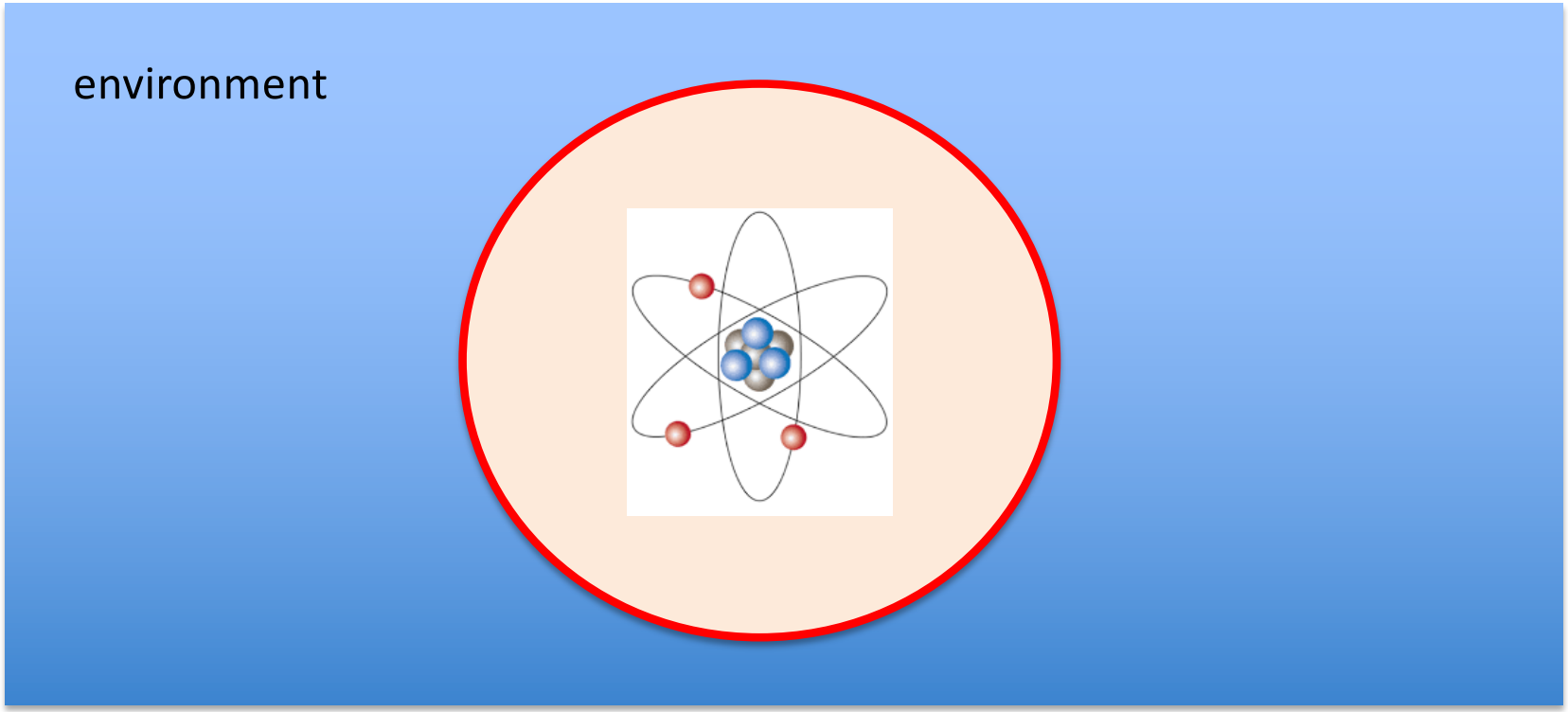
The quantum technology compromise



- photons from distant galaxies can remain in a pure polarization state for millions of years…
- is this useful? yes for "storage"

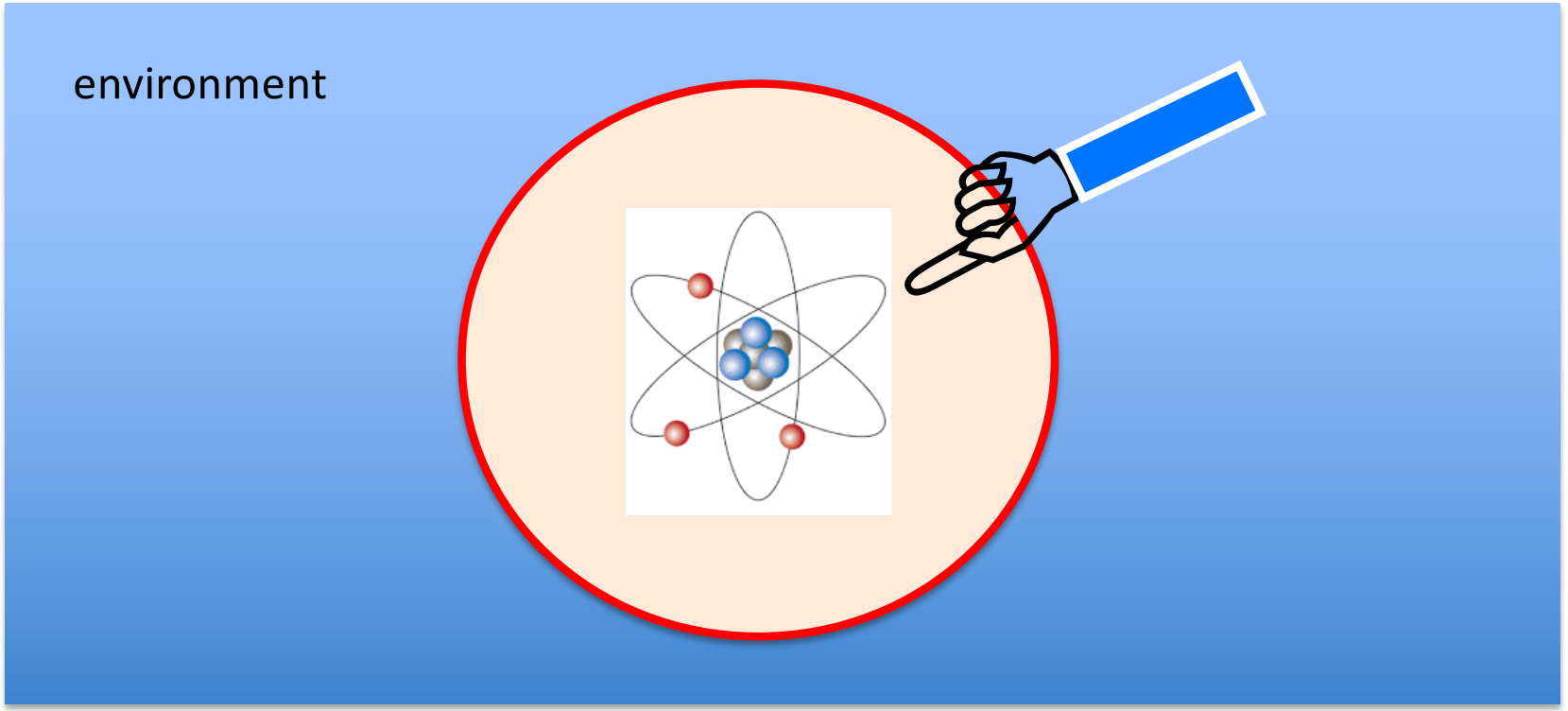
  no for manipulation + calculation.

The quantum technology compromise



- this is not an ideal situation!

The quantum technology compromise



- this is not an ideal situation!
- To implement quantum technologies we must be able to manipulate our quantum systems
- Inevitably this means the system must be open to some extent!

Summary

- quantum systems which interact with the environment are “open”
  - pure ensembles evolve into mixed ensembles
  - the energy of the system decays
  - the coherence of superpositions decays
- different systems interact with the environment in different ways
- quantum technology requires systems to be open
  - even if we could completely isolate our system this would not be useful for quantum technologies
  - to interact we must periodically open the system